

Figure

Redundancy Audit

A visual pass over every plot and photo in `results.tex` and `methodology.tex`, looking for the kind of duplication the R2-01 request named directly — repeated plot types and near-identical curves between the simulation and hardware campaigns. Read-only: nothing in the paper was changed to produce this.

91 images inventoried 3 candidate groups flagged
2 scenario figures recommended to cut Generated 2026-08-26

91 individual images across both sections	15 plots built from only 3 recurring panel types
7 sim/real pairs reviewed as validation evidence	18 photos across three 6-frame montage figures

GROUP A • HIGHEST PRIORITY

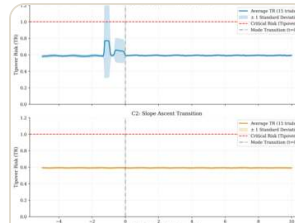
The same three panel types, drawn five times

The simulation's variable-topography battery established a 3-panel plot family — a Tipover-Risk profile, a switching-delay ECDF, and a Roll/Pitch RMS phase space. The physical multi-stimuli campaign (already condensed into Table `tab:consolidated_physical_metrics` this round) then redraws the same three panel *types* for all four stimulus scenarios. The numbers already live in the table — these are the figures that still repeat them visually.

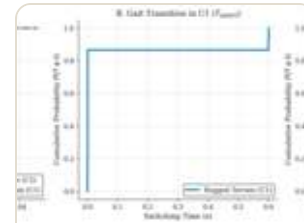
REFERENCE

Simulation quantitative battery (variable topography, 15 trials $\times$ 2 terrains)

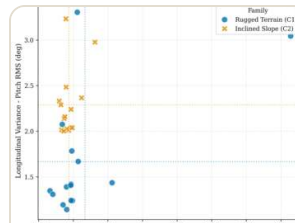
This is the plot family every physical-scenario figure below reuses: a Tipover-Risk profile, an ECDF of switching delay, and a Roll/Pitch RMS phase-space or range chart.



FigA – Tipover Risk (TR) profile



FigB – ECDF decision delay

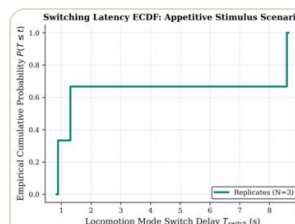


FigC – Roll/Pitch RMS phase space

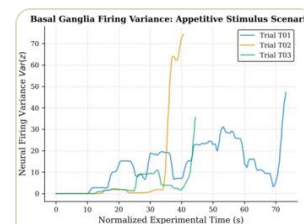
REVIEW – CANDIDATE TO CUT

Physical: Appetitive scenario

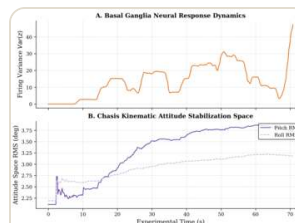
Panel shapes are near-identical to Aversive's (below) except the ECDF settles over a wider window; the underlying numbers already live in Table tab:consolidated_physical_met.



Appetitive – ECDF switching delay



Appetitive – Basal ganglia Var(z) dynamics

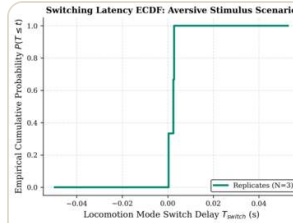


Appetitive – Roll/Pitch RMS

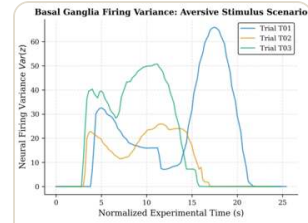
REPRESENTATIVE — KEEP

Physical: Aversive scenario

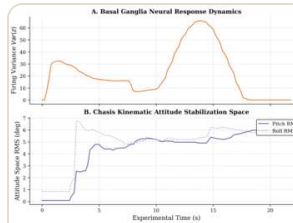
Clear instantaneous-reflex signature — the most visually distinct of the four; good candidate to keep as a representative figure.



Aversive — ECDF
switching delay



Aversive — Basal
ganglia Var(z)
dynamics

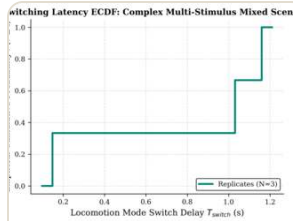


Aversive —
Roll/Pitch RMS

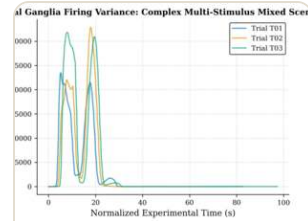
REPRESENTATIVE — KEEP

Physical: Complex scenario

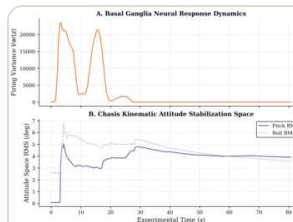
Three-phase Var(z) profile is the most information-dense of the four — good candidate to keep as the second representative figure.



Complex — ECDF
switching delay



Complex — Basal
ganglia Var(z)
dynamics

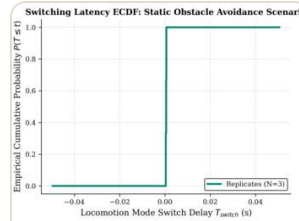


Complex — Roll/Pitch
RMS

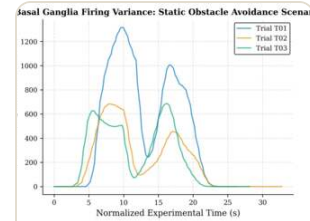
REVIEW — CANDIDATE TO CUT

Physical: Obstacle scenario

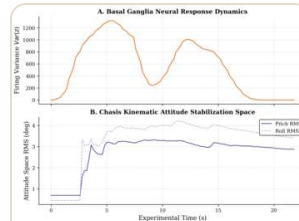
Panel shapes closely mirror Aversive's (near-instant ECDF, single transient spike, rhythmic RMS) — the weakest case for a dedicated 3-panel figure.



Obstacle — ECDF
switching delay



Obstacle — Basal
ganglia $\text{Var}(z)$
dynamics



Obstacle —
Roll/Pitch RMS

GROUP B • SIM VS. PHYSICAL

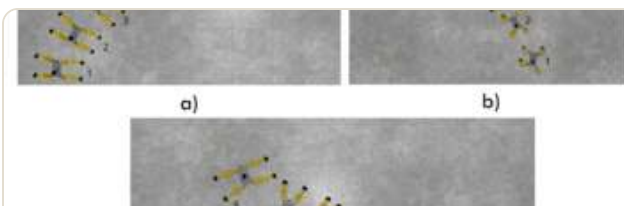
Paired trajectory & raster plots

These are not true duplicates — each pair shows the same experiment run on two different platforms, which is the actual sim-to-real validation evidence the paper is built on. Flagged here only where the pairing itself looks stretched thin across too many near-identical figures.

LEGITIMATE PAIR — KEEP

Single-stimulus trajectory

Core sim-to-real validation pair — the trajectory shapes are the direct evidence the controller transfers to hardware. Keep both.



Simulation

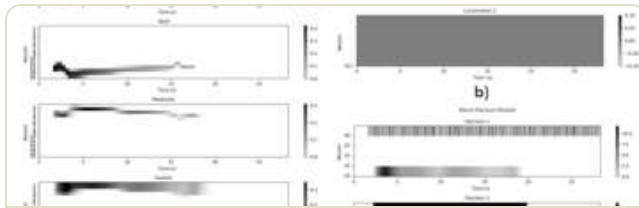


Physical

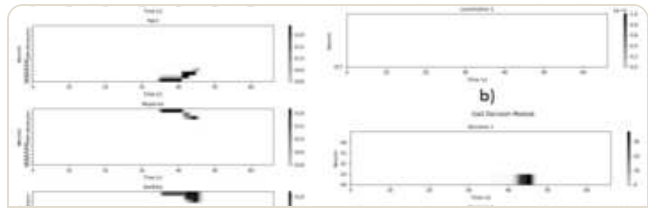
LEGITIMATE PAIR – KEEP

Obstacle-stimulus raster

Shows the same neural-activation story on both platforms; needed to support the sim → real claim.



Simulation

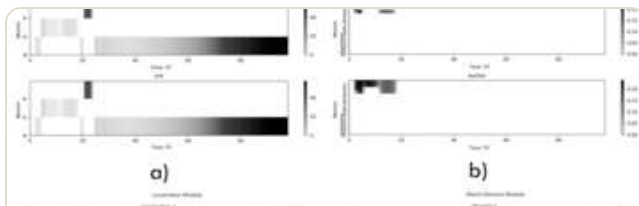


Physical

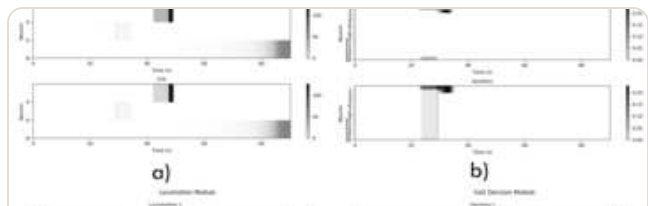
LEGITIMATE PAIR – KEEP

Multi-stimuli raster

Same reasoning as above, for the 3-stimuli scenario.



Simulation



Physical

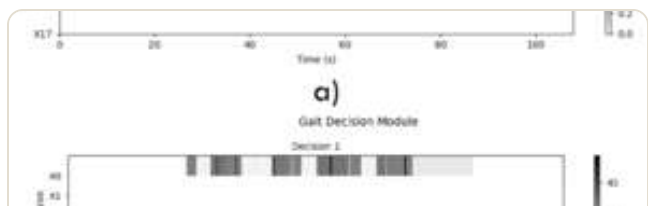
WORTH A SECOND LOOK

Rough-terrain neural activations

Legitimate pair, but visually near-identical in layout to the incline-terrain pair below — four raster figures for two terrain types. Consider whether two of the four could merge into one multi-panel figure.



Simulation

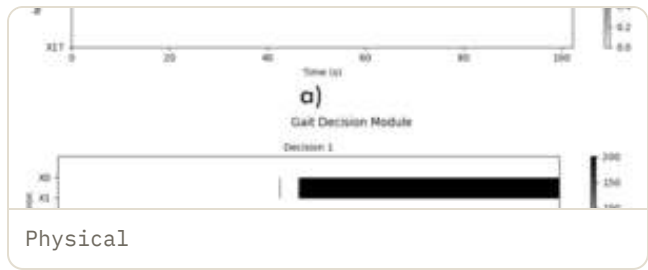
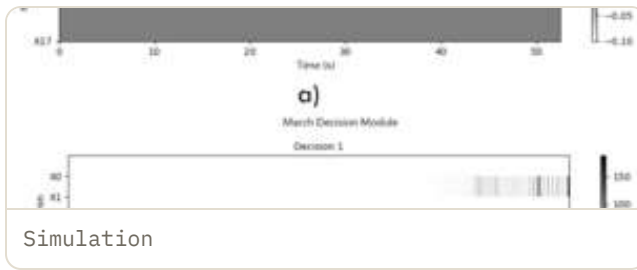


Physical

WORTH A SECOND LOOK

Incline-terrain neural activations

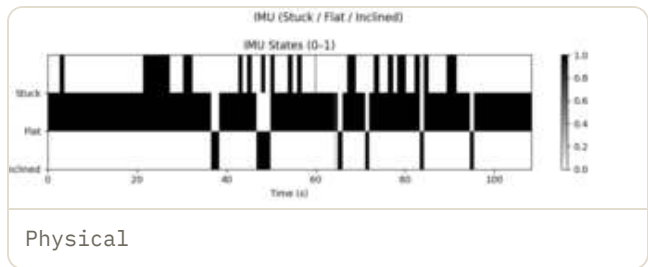
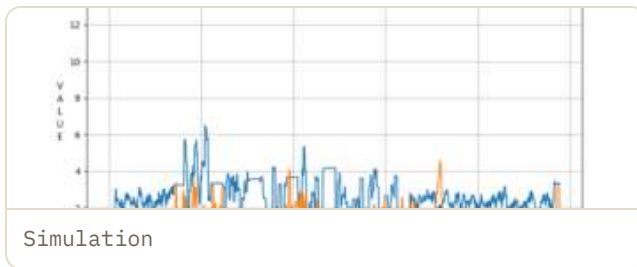
See note on the rough-terrain pair above — same layout, different terrain.



LEGITIMATE PAIR — KEEP

Rough terrain — accel std-dev vs. MLP timeline

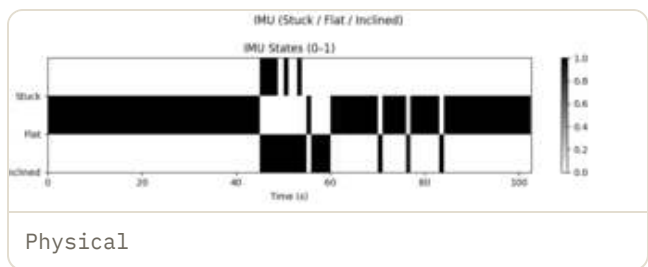
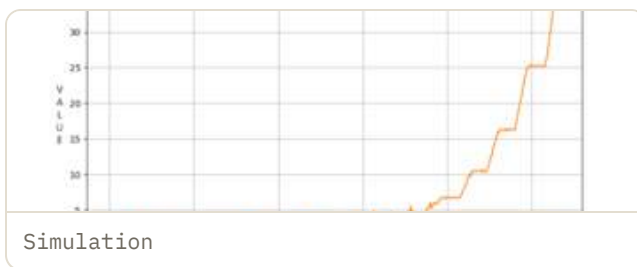
Despite the parallel caption style, sim plots raw acceleration std-dev while real plots MLP output activations — different content, not a duplicate.



LEGITIMATE PAIR — KEEP

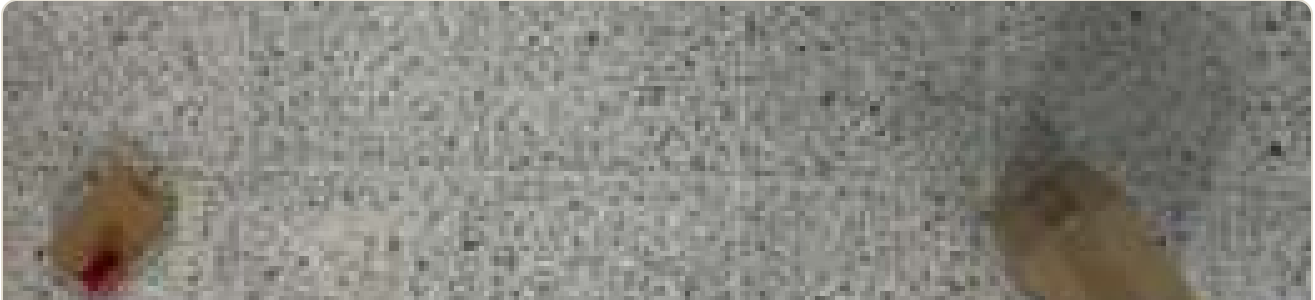
Incline terrain — pitch angle vs. MLP timeline

Same distinction as above — keep both.



Three 6-frame photo montages

Not plot duplication in the strict sense, but the same full-width, six-photo grid format used three times in a row for three different physical trials — worth a look for whether all eighteen frames earn their page space.



1 of 6 — Multi-stimuli response (D1–D6)

results.tex, physical multi-stimuli test — full-width figure, six frames.



1 of 6 — Irregular terrain traversal (E1–E6)

results.tex, physical rough-terrain test — full-width figure, six frames.



1 of 6 — Compound terrain course (Parte 1–6)

results.tex, physical compound-terrain test, 15 trials — full-width figure, six frames.

REVIEWED, NO ACTION NEEDED

Everything else that looked similar at a glance

Checked because their captions or filenames rhymed with something else in the paper — each turned out to show genuinely distinct content.

Modos_Locomocion / Modos_LocomocionV2	Two distinct gait-decision network architectures — the original IMU-threshold version and the MLP-informed replacement. Both are referenced and needed.
NeuLidarFB / NeuLidarLR	Front–rear and left–right views of the same LiDAR encoding ring — complementary angles, not a repeat.
fig1_confusion_matrix ... fig6_computational_cost	Six MLP-evaluation figures (confusion matrix, ROC, training loss, architecture ablation, classifier comparison, latency) — each reports a different metric, no overlap.
estimulo_apetente(_real) / estimulo_adversivo(_real)	Sim stimulus design vs. the physical object used in hardware trials — necessarily paired, not duplicated.
TERRENO_IRREGULAR / TERRENO_INCLINADO / TERR_IMU	Three different physical/simulated terrains, not variants of one terrain.
RES_EST_MUL / NEU_EST_MUL (sim)	Trajectory vs. raster view of the same trial — standard complementary pairing used consistently across every stimulus scenario in the paper.

REFERENCE

Full figure inventory

Every `\includegraphics` call in both files, grouped by the subsection it appears in, for cross-checking against the groups above.

—results.tex — Simulation Results (24 images)

FILE	CONTEXT
Depot_simulation.jpeg	Warehouse-inspired Gazebo environment
RES_EST_UNI.png	Single-stimulus trajectories (sim)
NEU_EST_UNIG.png / NEU_EST_UNIB.png	Single-stimulus raster plots (sim, obstacle / appetitive)
fig1/fig2_{Appetitive,Aversive,Obstacle,Complex}_*.png	Quantitative validation, 4 scenarios × 2 panels (sim)
NEU_EST_MUL.png / RES_EST_MUL.png	Multi-stimuli raster & trajectory (sim)
RES_IMU.png / NEU_IMU.png / GRAPH_IMU.png	Rough-terrain response (sim)
RES_IMU2.0.png / NEU_IMU2.png / GRAPH_IMU2.png	Incline-terrain response (sim)
FigA/B/C_*.png	Quantitative variable-topography battery (sim) — see Group A
fig_stability_phases.png	Six-phase crawl-gait cycle diagram
fig_transition_phases_v9_part1/2.png	Six mode-transition frame sequences

—results.tex — Physical Implementation Results (35 images)

FILE	CONTEXT
RES_EST_UNI_real.png / NEU_EST_UNIG_real.png	Single-stimulus trajectory & raster (physical)
NEU_EST_MUL_real.png	Multi-stimuli raster (physical)
D1-D6.png	Multi-stimuli 6-frame photo montage — see Group C
fig1-fig6_*.png	MLP terrain-classifier evaluation (6 distinct metrics)
NEU_IMU_real.png / GRAPH_IMU_real.png	Rough-terrain response (physical)
E1-E6.png	Irregular-terrain 6-frame photo montage — see Group C
RES_IMU2.0_real.png / NEU_IMU2_real.png / GRAPH_IMU2_real.png	Incline-terrain response (physical)
Prueba_Terreno_Parte_1-6.png	Compound-terrain 6-frame photo montage — see Group C
{Appetitive,Aversive,Complex,Obstacle}_Real_Plot_1/2/3_*.png	Quantitative multi-stimuli battery, 4 scenarios × 3 panels — see Group A
FigReal_Morphological_Transition_Energy.png	Energy consumption across mode transitions (physical only, no sim equivalent)
FigReal_Terrain_Latencies_4Panels.png	Four-panel neural/electromechanic latency breakdown



—methodology.tex — all subsections (19 images)

FILE	CONTEXT
robot.png / pata.png	Platform & limb 3D design
TERR_IMU.PNG	Simulated terrain model (3 topologies)
estimulo_apetente.png / estimulo_adversivo.png	Simulated stimulus objects
NeuLidarFB.png / NeuLidarLR.png / IntegrationLidar.png	LiDAR encoding ring & integration network
ganglios.png	Basal Ganglia module diagram
Decisiones_Locomotoras.png	Locomotion decision module diagram
Modos_Locomocion.png / Modos_LocomocionV2.png	Gait-decision network, pre/post MLP integration
MLP.PNG / VENTANA_DATOS.png	MLP structure & input-window diagram
PETER.jpeg	Assembled physical robot
estimulo_apetente_real.jpeg / estimulo_adversivo_real.jpeg	Physical stimulus objects
TERRENO_IRREGULAR.jpeg / TERRENO_INCLINADO.jpeg	Physical test terrains

How this was built: every `\includegraphics` in `sections/results.tex` and `sections/methodology.tex` was catalogued, grouped by the subsection and experimental scenario it belongs to, then cross-checked against its caption text for structural repeats (same panel types, same 6-frame format, same sim/real pairing pattern). Thumbnails are downsampled copies of the real assets in `assets/`, for review only. No section file, patch, or `PROGRESS.md` row was touched.